Utilisation of Funds & Internal Transfer Pricing — Why We Built It This Way

`docs/ftp_charges_credits_realised_margin.md` walks through *how* we turn a rate and a balance into an actual charge, credit, and realised margin. This is the companion piece — *why* we built FTP as an internal transfer-pricing mechanism in the first place, and why the specific design choices inside it sit where they do, across the maths, the financial logic, the software design, and — most of all here — the funding and profitability strategy it exists to serve.

Decision: FTP as an internal market for funds, not a top-down allocation

Mathematically — every account gets its own charge or credit, computed from its own balance and its own curve-implied rate at its own WAL, rather than a pooled cost of funds allocated proportionally across the book. That's a meaningfully different calculation: it prices each unit of funding at what it would actually cost (or earn) on the curve at that specific tenor, not at an average.

Financially — a top-down allocation hides which specific products and tenors are actually driving funding cost or benefit; a bottom-up, per-account charge doesn't. It's the difference between knowing your average cost of funds and knowing exactly which parts of the book are expensive to fund and which parts are funding you. A survey of large-bank practice found this distinction was, in practice, the single biggest gap between weak and strong transfer-pricing implementations: institutions that priced liquidity at one average rate rather than attributing it by product and tenor systematically mispriced both ends of the balance sheet (Grant, 2011).

In the software — we compute `BASE_CHARGE`, `LP_CHARGE`, and `FTP_CHARGE` as three separate accrual figures per account (or tranche), not just one blended number, specifically so the consolidation layer can report *where* a period's total FTP charge came from — term cost versus liquidity cost — not just the total.

As a business decision — this is the whole point of the exercise: FTP exists to answer "is this product, this branch, this segment actually creating value, once it's honestly charged for the funding it uses or credited for the funding it provides." A pooled or averaged approach can't answer that question; a genuine internal transfer price, computed account by account, can.

Decision: designing it as a zero-sum, double-entry mechanism

Mathematically — Treasury is modelled as the mirror counterparty to every business-side charge or credit: what one side pays, the other receives. A perfectly matched book — assets and liabilities of identical size and tenor — nets to exactly zero by construction, not by coincidence.

Financially — this is what makes a non-zero net position *meaningful* rather than just noise. A real, non-zero net position reflects a genuine, quantifiable decision the bank has made — funding longer-dated assets with shorter-dated liabilities, or vice versa — the actual, priced cost (or benefit) of maturity transformation, which is one of a bank's core structural sources of profit or risk.

In the software — every account's charge and every account's credit are computed from the same curve and the same formulas, deliberately symmetric, rather than assets and liabilities running through separate pricing logic that could drift out of consistency with each other over time.

As a business decision — this is the mechanism that turns "how are we funding ourselves and what's it worth" into a number management can actually act on. A business unit funding itself expensively relative to its own asset yield shows up directly as compressed realised margin; a business unit sitting on cheap, stable funding shows up as a genuine funding advantage the bank can choose to lean into — deliberately grow that book, price more competitively, or redeploy that advantage elsewhere. That's the strategic value of the whole exercise: it turns funding structure into a visible, decidable input to profitability strategy, not a background fact — and it's the same value a large-scale "sticky" deposit base represents in the non-maturing liability literature more broadly (Kalkbrener & Willing, 2004): a franchise advantage that's only usable strategically once it's actually quantified.

Decision: recomputing weighted averages at every rollup level, not averaging the daily figures

Mathematically — a weighted average recomputed from a period's own totals (`Σ(rate × balance) / Σ(balance)`) is not the same number as an average of daily weighted averages, once balances move within the period. Averaging averages introduces a real bias whenever balance size and rate aren't independent of each other across days — which, in practice, they usually aren't.

Financially — getting this wrong doesn't just introduce noise, it introduces a directional bias into every rolled-up margin figure a business reviews above the daily level.

In the software — we recompute weighted rates (and the pricing gap derived from them) fresh at every consolidation level from that level's own underlying totals, rather than propagating an already-averaged figure upward.

As a business decision — decisions get made off weekly, monthly, and segment-level numbers far more often than off daily ones. A systematic bias at exactly the level most decisions actually happen at is the worst place for an error like this to live.

Decision: a materiality tolerance for calling "net neutral," not exact zero

Mathematically — in any real system, floating-point accrual arithmetic and genuine rounding mean an exactly-zero net position is essentially never observed, even for a book that's economically matched. A tolerance band is what lets us distinguish "genuinely neutral" from "genuinely a net payer or receiver" without over-reading noise.

Financially/business-wise — this is what makes the transfer- direction reporting actually usable for strategy. Knowing whether a branch or segment is structurally a net *user* of internal funding or a net *provider* of it — reliably, not on a coin-flip driven by rounding — is core "utilisation of funds" information: it tells management where the bank's actual funding surplus and funding need sit across the organisation, which is a direct input to where to grow lending, where to compete harder for deposits, and where the bank's structural funding advantages and gaps actually are.

See also `docs/ftp_charges_credits_realised_margin.md`.

References

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Kalkbrener, M., & Willing, J. (2004). Risk management of non-maturing liabilities. *Journal of Banking & Finance, 28*(7), 1547–1568.